CSC-125 Python Programming
Midterm Programming Practice - Solutions

Covers Weeks 1-9: Variables, Data Types, Strings, Operators, Conditionals, Lists, Tuples, and Loops

This document contains complete solutions to all programming problems.

Note: Some problems may have multiple correct solutions. The answers provided here are example solutions that meet all requirements.

Part 1: Variables and Basic Operations (Weeks 1-4)

1. Temperature Converter

Write a program that converts a temperature from Fahrenheit to Celsius. Your program should:

- Ask the user to enter a temperature in Fahrenheit
- Convert it to Celsius using the formula: C = (F - 32) * 5/9
- Print the result with a descriptive message

Hint: Remember to convert the input to a float!

ANSWER:

```
fahrenheit = float(input("Enter temperature in Fahrenheit: "))
celsius = (fahrenheit - 32) * 5/9
print(f"Temperature in Celsius: {celsius}")
```

Alternative: You could also use print with commas: `print("Temperature in Celsius:", celsius)`

2. Age Calculator

Write a program that calculates how old someone will be in 10 years. Your program should:

- Ask the user for their current age
- Calculate their age in 10 years
- Print a message showing their future age

Hint: Use int() to convert the input to an integer

ANSWER:

```
current_age = int(input("Enter your current age: "))
future_age = current_age + 10
print(f"In 10 years, you will be {future_age} years old")
```

Alternative: You could also use print with commas: `print("In 10 years, you will be", future_age, "years old")`

3. Rectangle Area

Write a program that calculates the area of a rectangle. Your program should:

- Ask the user for the length of the rectangle
- Ask the user for the width of the rectangle
- Calculate the area (length * width)
- Print the result

ANSWER:

```
length = float(input("Enter the length: "))
width = float(input("Enter the width: "))
area = length * width
print(f"The area is {area}")
```

Alternative: You could also use print with commas: `print("The area is", area)`

Part 2: Strings (Week 3)

4. Name Formatter

Write a program that formats a person's name. Your program should:

- Ask the user for their first name
- Ask the user for their last name
- Print their full name in the format: "Last, First" (all uppercase)

Hint: Use the .upper() method

ANSWER:

```
first_name = input("Enter your first name: ")
last_name = input("Enter your last name: ")
formatted_name = last_name + ", " + first_name
print(formatted_name.upper())
```

Alternative: Could also convert each name to uppercase first: `print(last_name.upper() + ", " + first_name.upper())`

5. Initials

Write a program that gets a person's initials. Your program should:

- Ask the user for their first name
- Ask the user for their last name
- Print their initials (first letter of each name, uppercase)

Hint: Use indexing [0] and .upper()

ANSWER:

```
first_name = input("Enter your first name: ")
last_name = input("Enter your last name: ")
initials = first_name[0].upper() + last_name[0].upper()
print(f"Your initials are: {initials}")
```

Alternative: You could also use print with commas: `print("Your initials are:", initials)`

Part 3: Conditionals (Week 5)

6. Grade Checker

Write a program that determines if a student passed or failed. Your program should:

- Ask the user for a test score
- If the score is 60 or above, print "Pass"
- Otherwise, print "Fail"

ANSWER:

```
score = int(input("Enter your test score: "))
if score >= 60:
    print("Pass")
else:
    print("Fail")
```

Note: Could also use float() if decimal scores are allowed.

7. Even or Odd

Write a program that determines if a number is even or odd. Your program should:

- Ask the user for a number
- If the number is even, print "Even"
- If the number is odd, print "Odd"

Hint: Use the modulo operator (%) to check if a number is divisible by 2

ANSWER:

```
number = int(input("Enter a number: "))
if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

8. Letter Grade

Write a program that converts a numeric score to a letter grade. Your program should:

- Ask the user for a score (0-100)
- Print "A" if score >= 90
- Print "B" if score >= 80
- Print "C" if score >= 70
- Print "D" if score >= 60
- Otherwise print "F"

Hint: Use elif statements

ANSWER:

```
score = int(input("Enter your score: "))
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
elif score >= 60:
    print("D")
else:
    print("F")
```

Important: Order matters! Must check from highest to lowest score.

Part 4: Lists (Weeks 6-7)

9. List Statistics

Write a program that analyzes a list of numbers. Your program should:

- Create a list: numbers = [10, 25, 30, 15, 40]
- Print the length of the list
- Print the maximum value in the list
- Print the minimum value in the list

Hint: Use len(), max(), and min() functions

ANSWER:

```
numbers = [10, 25, 30, 15, 40]
print(f"Length: {len(numbers)}")
print(f"Maximum: {max(numbers)}")
print(f"Minimum: {min(numbers)}")
```

Alternative: You could also use print with commas: `print("Length:", len(numbers))`

10. Shopping List

Write a program that manages a shopping list. Your program should:

- Create an empty list called shopping_list
- Add "milk" to the list
- Add "bread" to the list
- Add "eggs" to the list
- Print the complete shopping list
- Print how many items are on the list

Hint: Use .append() method

ANSWER:

```
shopping_list = []
shopping_list.append("milk")
shopping_list.append("bread")
shopping_list.append("eggs")
print(f"Shopping list: {shopping_list}")
print(f"Number of items: {len(shopping_list)}")
```

Alternative: You could also use print with commas: `print("Shopping list:", shopping_list)`

Part 5: For Loops (Week 8)

11. Countdown

Write a program that counts down from 5 to 1. Your program should:

- Use a for loop to print numbers from 5 down to 1
- Print "Blastoff!" after the countdown

Hint: Use range(5, 0, -1)

ANSWER:

```
for i in range(5, 0, -1):
    print(i)
print("Blastoff!")
```

Note: Make sure "Blastoff!" is NOT indented - it should print after the loop ends.

12. Sum Calculator

Write a program that calculates the sum of numbers from 1 to 10. Your program should:

- Create a variable called total and set it to 0
- Use a for loop to add each number from 1 to 10 to total
- Print the final total

Hint: Use range(1, 11)

ANSWER:

```
total = 0
for i in range(1, 11):
    total = total + i
print(f"Total: {total}")
```

Alternative: You could use `total += i` instead of `total = total + i`, or use print with commas

13. List Doubler

Write a program that doubles each number in a list. Your program should:

- Create a list: numbers = [2, 4, 6, 8]
- Use a for loop to print each number multiplied by 2

ANSWER:

```
numbers = [2, 4, 6, 8]
for num in numbers:
    print(num * 2)
```

Alternative: Could also use indexed loop: `for i in range(len(numbers)): print(numbers[i] * 2)`

Part 6: While Loops (Week 9)

14. Password Checker

Write a program that keeps asking for a password until correct. Your program should:

- Set a correct password (you choose the password)
- Use a while loop to keep asking the user for the password
- If the password is correct, print "Access granted!" and exit the loop
- If incorrect, print "Try again" and ask again

Hint: The loop should continue while the password is incorrect

ANSWER:

```
correct_password = "python123"
password = input("Enter password: ")
while password != correct_password:
    print("Try again")
    password = input("Enter password: ")
print("Access granted!")
```

Alternative Solution: Could also use a flag variable or `while True:` with `break`

15. Number Guesser

Write a program that counts how many guesses it takes. Your program should:

- Set a secret number (you choose the number)
- Create a counter variable starting at 0
- Use a while loop to keep asking the user to guess
- Increment the counter each time they guess
- When they guess correctly, print how many guesses it took

ANSWER:

```
secret_number = 42
counter = 0
guess = int(input("Guess the number: "))
counter = counter + 1
while guess != secret_number:
    guess = int(input("Try again: "))
    counter = counter + 1
print(f"Correct! It took you {counter} guesses")
```

Alternative: You could use print with commas: `print("Correct! It took you", counter, "guesses")` or use `counter += 1`